

Balancing Pedestrian Service and Robot Access: An Interpretable Admission Framework for Sidewalk Delivery Robots

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ABSTRACT

Operating permission for sidewalk delivery robots leaves a local allocation question: how much robot entry flow can a sidewalk accommodate under a stated pedestrian-service objective? We develop an admission framework that checks applicability and pedestrian baseline service before estimating a candidate entry rate from width and demand. City-held-out evaluation compares conservative allocation with alternative estimators. All positive recommendations were tested directly at their specified flows. Across 164 scenarios, the margin-based method offered 98 positive recommendations, with 98 passes and 0 failures. Removing the margin increased access to 136 positive recommendations, with 118 passes and 18 failures. Median reference utilization increased from 16.7% to 75.0%. Of the 38 additional opportunities, 32 passed and 6 failed, showing the access cost of conservative allocation. An unconstrained width–demand estimator produced similar aggregate outcomes to the margin-based method. Transfer to 13 original and 5 entrance-repaired Hong Kong configurations extended the analysis to irregular polygons. Geometry, robot size, and flow composition affected applicability. The framework supports service-aware evaluation of robot access under specified operating conditions.

INTRODUCTION

U.S. states including Virginia and Washington permit sidewalk delivery devices under specified operating conditions (Virginia General Assembly 2020; Washington State Legislature 2019). Singapore has enabled supervised public-path trials through regulatory exemptions (Land Transport Authority 2022; Infocomm Media Development Authority 2021). These arrangements address operation and pedestrian protection; the District of Columbia also limits fleet size in initial permits (District Department of

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Transportation 2026). Yet operating permission and fleet limits do not determine how many robots per minute should enter a particular sidewalk. That decision depends on local width, pedestrian demand, and the service that walking should retain.

Delivery robots offer potential freight benefits (Jennings and Figliozzi 2019), but each movement shares space primarily used by pedestrians. Research has examined robot regulation and deployment (Kovacica et al. 2024), pedestrian interactions (Gehrke et al. 2022), and delivery routing under uncertain travel conditions (Tong and Simoni 2026). These studies motivate a related traffic-engineering question: how much robot access should be offered on a defined sidewalk under a stated pedestrian-service objective? The decision variable here is a segment entry rate, measured in robots/min.

A regression estimate alone leaves several decisions unresolved. Some geometries lie outside the assessed domain. Some pedestrian-only conditions already fail the service criterion. Other conditions support a positive robot recommendation, but the amount offered depends on how conservatively the available space is allocated. These cases call for different responses: further assessment, zero admission, or a candidate entry rate linked to a service test.

We develop an interpretable framework that connects these responses through applicability checks, a pedestrian-service gate, and a width–demand estimate with a margin. City-held-out comparisons quantify how conservativeness changes robot access and pedestrian-service outcomes. Tests with frozen development parameters examine transfer to Hong Kong sidewalk polygons. The contributions are an explicit admission decision structure, a direct comparison of access and service at recommended flows, and an assessment of the conditions that limit geometric transfer.

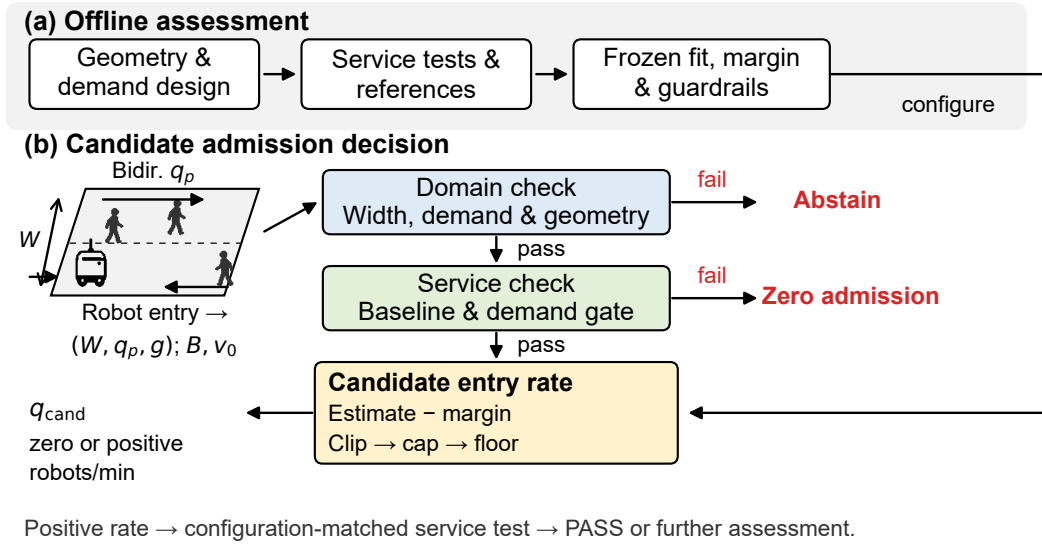


Fig. 1. Admission decision structure. Applicability determines whether a scenario can be assessed. Pedestrian baseline and demand checks precede candidate-rate calculation. A positive candidate proceeds to a service test at the specified configuration and flow.

FROM PEDESTRIAN SERVICE TO ROBOT ACCESS

Study Scenarios

The study contains 400 geometry–demand scenarios from 140 sidewalk cells in Amsterdam, Melbourne, New Taipei, New York, Seattle, Taipei, and Taoyuan. GIS-derived widths and centreline descriptors define idealized corridors of nominal length 50 m. The 280 main scenarios use demand priors informed by nearby residential, commercial, tourism, and transit activity. Another 120 scenarios extend the demand range. Demand is specified for the experiments rather than measured as pedestrian arrivals in these cities.

Simulations use SUMO 1.27.1 with JuPedSim (Lopez et al. 2018). Robots are represented as walking agents with dimensions 0.96 by 0.7 m, interaction radius 0.48 m, and nominal speed 1.39 m/s. Pedestrians have radius 0.25 m. Their desired speeds have mean 1.25 m/s and standard deviation 0.15 m/s, bounded by 0.8 and 1.8 m/s. Pedestrian demand is divided equally between directions, and robots enter in one direction. Regular scheduled arrivals specify demand; random seeds vary stochastic agent properties.

Each run uses a 0.1 s time step, a 100 s warm-up, and a 240 s measurement window. A further 100 s allows trips to finish and is excluded from service measurement. A central 20 m zone defines the speed and density measurements. In irregular polygons, positions are projected onto the measurement axis. Zone length multiplied by representative width supplies the density denominator.

Physical feasibility is checked before service evaluation. Eroding the walkable polygon by the robot radius leaves no feasible centre region in 44 scenarios. A further 5 pedestrian-only runs exceed the 300 s wall-clock limit and lack completed measurements. These two groups are excluded rather than assigned zero admission. Completed runs with congestion retain their adverse service outcomes.

Pedestrian-Service Criterion

The service objective combines pedestrian speed retention, density, and a completed-trip flow ratio. For seed s , mean speed $\bar{v}_s$ averages pedestrian observations in the measurement zone and window. Mean density $\bar{k}_s$ averages the pedestrian count divided by zone area. Robots are excluded from both measures. Baseline speed v_0 is the mean of the pedestrian-only seed means for the same scenario and configuration. A mixed-flow seed qualifies when

$$\bar{v}_s/v_0 \geq 0.90, \quad \bar{k}_s \leq 1.20, \quad 0.90 \leq T_s \leq 1.20. \quad (1)$$

Density is in ped/m². The thresholds express the study’s service objective. Speed retention compares each mixed-flow seed with a common baseline mean.

The flow ratio is $T_s = N_{\text{arr},s}^{\text{comp}} / N_{\text{dep},s}^{\text{comp}}$. Its numerator counts completed pedestrian records arriving during measurement; its denominator counts completed records departing during measurement. When the denominator is zero, T_s is undefined and the seed fails this component. Scheduled agents, first observations, completed trips, and window stocks are retained as separate demand diagnostics. First observation establishes visibility to the simulator, not an exact insertion time into the walking model.

A flow qualifies when at least 29 of 30 seeds satisfy Eq. 1. Pedestrian-only qualification B applies the density and flow-ratio criteria with the same acceptance count.

All primary references use seeds 0–29. Independent seeds 10000–10029 are used only for replication, with a separate baseline mean.

The simulation reference q^* is the highest qualifying flow in the reference scan. It is zero when no tested flow qualifies and is censored at the 20 robots/min ceiling. Coarse design flows are 1, 2, 3, 4, 5, 6, 8, 10, 12, 15, 18, 20 robots/min; higher levels and integer refinements are applied selectively. Scan support can differ among scenarios. Later recommendation tests are evaluated separately from reference construction. The resulting sample contains 185 positive uncensored references, 154 zeros, and 12 ceiling-censored references.

Admission Decision

Figure 1 connects service evaluation to robot access. Inputs are clear width W , pedestrian demand q_p , geometry descriptors g , baseline qualification B , and baseline speed v_0 . Applicability is assessed first. Missing inputs, invalid evidence, or a scenario outside the assessed domain produce abstention.

The domain $\mathcal{D}$ covers widths of 1.6–3 m and positive pedestrian demand no greater than 60 ped/min. Specific demand is $x = q_p/W < 33.33$ ped/min/m. Straight and mildly curved geometries are included up to sinuosity 1.05. Concentrated turns are excluded when maximum and cumulative angles both reach 20 degrees and their ratio reaches 0.6.

For an assessable scenario, the pedestrian-service gate G requires $B = 1$, $v_0 \geq 0.8$ m/s, and $q_p < C(W)$. Failure of this gate returns zero admission. Otherwise, the method subtracts a margin from the nominal estimate, applies flow ceilings, and rounds down:

$$q_{\text{cand}} = \begin{cases} \text{abstain,} & \text{inputs invalid or } (W, q_p, g) \notin \mathcal{D}, \\ 0, & (W, q_p, g) \in \mathcal{D}, G = 0, \\ \lfloor \min\{q_{\text{max}}, Q_{\text{low}}(W), [q_{\text{nom}} - \Delta]_+\} \rfloor, & (W, q_p, g) \in \mathcal{D}, G = 1. \end{cases} \quad (2)$$

Here $[z]_+ = \max(0, z)$ and $q_{\text{max}} = 20$ robots/min. Margin subtraction and rounding may also produce zero. This distinguishes a pedestrian-service gate from a conservative allocation decision. A positive output is a candidate entry rate for configuration-matched testing.

The margin-based method M0 uses the width–demand estimate

$$q_{\text{nom}} = cW(q_p/W)^p, \quad \log(q^*/W) = \log c + p \log x + \varepsilon. \quad (3)$$

Ordinary least squares fits positive uncensored training references. Zero and ceiling-censored references are omitted from this logarithmic fit but retained in margin calibration. The margin Δ is the 0.80 quantile of $[q_{\text{nom}} - q^*]_+$ across training scenarios with defined references. Fitting and margin calibration use the recorded rounded x values. Candidate-rate calculation evaluates q_p/W from the stored inputs.

The demand gate $C(W)$ and low-demand ceiling $Q_{\text{low}}(W)$ were established from the synthetic grid and held fixed (Table 1). They supplement the fitted relation by limiting the range of offered access.

Table 1. Demand gate and low-demand ceiling, interpolated linearly between width knots.

W (m)	1.6	1.7	1.8	2.1	2.4	2.7	3
$C(W)$ (ped/min)	35.0	35.0	45.0	45.0	55.0	90.0	90.0
$Q_{\text{low}}(W)$ (robots/min)	5.0	6.0	8.0	10.0	18.0	20.0	20.0

EVALUATION DESIGN

City-Held-Out Comparison

Four methods share the same applicability checks, service gate, and ceilings. M1 removes the margin from M0, setting $\Delta = 0$. It isolates the effect of conservative allocation within this framework. M2 uses a width-only relation, $\log q_{\text{nom}} = a + b \log W$, with its own calibrated margin. M3 uses the unconstrained relation $\log q_{\text{nom}} = a + b \log W + d \log q_p$, also with its own margin. M0 imposes $b + d = 1$ in this representation. M0 has two fitted coefficients and M3 has three.

Each of the 7 folds fits coefficients and margin on 6 cities and predicts the remaining city. All cells from the held-out city are excluded from fitting. The previously selected gates and ceilings remain fixed across folds. Applicability screening leaves 164 scenarios from 64 unique cells, or 41.0% of the candidate sample. The 236 abstentions comprise 44 physical exclusions, 5 technical exclusions, and 187 operational-domain exclusions. Zero recommendations remain in the evaluation population.

Direct Recommendation Tests

The primary outcomes are robot access and pedestrian-service outcomes at the recommended flow. We report positive and zero recommendations on the common scenario set. Positive flows are matched to tests by scenario, entrance configuration, flow, and original seed block. Methods recommending the same flow share its test evidence. A complete 30-seed group produces PASS or FAIL according to Eq. 1. All positive recommendations have complete configuration-matched tests. Zero admission is not counted as a service pass.

Mean positive candidate flow describes the amount of access offered in robots/min. It is a nominal recommendation, separate from completed robot throughput. To describe pedestrian service continuously, we also summarize seed-level speed retention within tested groups.

Reference exceedance counts recommendations with $q_{\text{cand}} > q^*$. Mean absolute error (MAE) compares candidate rates with q^* on all 164 evaluated scenarios. Reference utilization is

$$U = \text{median}_{i: q_i^* > 0} (q_{\text{cand}, i} / q_i^*) . \quad (4)$$

Its denominator includes 139 scenarios with positive references, including 8 ceiling-censored references. Zero recommendations remain in the median, and ratios above unity are not clipped. Direct service tests and reference comparisons answer different questions: a recommendation can lie below q^* yet fail at its own tested flow.

Uncertainty intervals use 5000 paired cell-cluster resamples with replacement. Predictions, margins, and references remain fixed. The 95% percentile intervals describe variation in site composition; they exclude retraining and simulation-seed uncertainty.

Geometry Transfer to Hong Kong

The transfer experiment applies development parameters to irregular sidewalk polygons. Fitting on the full development set gives M0 parameters $c = 23.842$, $p = -0.930$, and $\Delta = 3.376$ robots/min. These parameters and the other methods' margins are frozen before the reported transfer calculations. Hong Kong data contribute to neither coefficient fitting nor margin selection.

The sample contains 32 candidate polygons. Domain screening retains 18 sites, of which 13 use original entrance configurations. The remaining 5 use entrance repairs within the original polygon. Repair moves entrance anchors to the nearest feasible points after erosion by 0.75 m and aligns the entrance stubs with the principal axis. Polygon shape, representative width, measurement axis, agent properties, and demand are retained. Each polygon contributes one final configuration. Original and repaired configurations are evaluated separately with matching baselines.

RESULTS

Conservativeness Trades Robot Access for Pedestrian Service

Removing the margin offered robot access in more scenarios and produced more failed service tests (Fig. 2; Table 2). M0 issued 98 positive recommendations and 66 zeros. Its positive recommendations included 98 passes and 0 failures. M1 issued 136 positives and 28 zeros, with 118 passes and 18 failures. Positive access increased by 38 scenarios when the margin was removed.

Table 2. City-held-out admission outcomes on 164 scenarios from 64 cells.

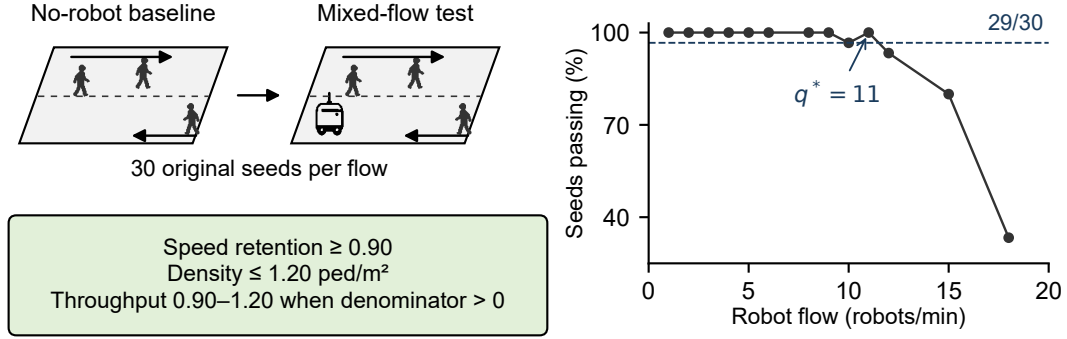
Method	Positive	Zero	PASS	FAIL	Exceed	MAE	U (%)
M0	98	66	98	0	0	4.54	16.7
M1	136	28	118	18	16	2.43	75.0
M2	59	105	59	0	0	6.98	0.0
M3	98	66	96	2	2	4.55	18.2

PASS and FAIL partition positive recommendations. Exceedance and MAE use all 164 scenarios; MAE is in robots/min. Utilization U uses 139 positive references, including zero recommendations.

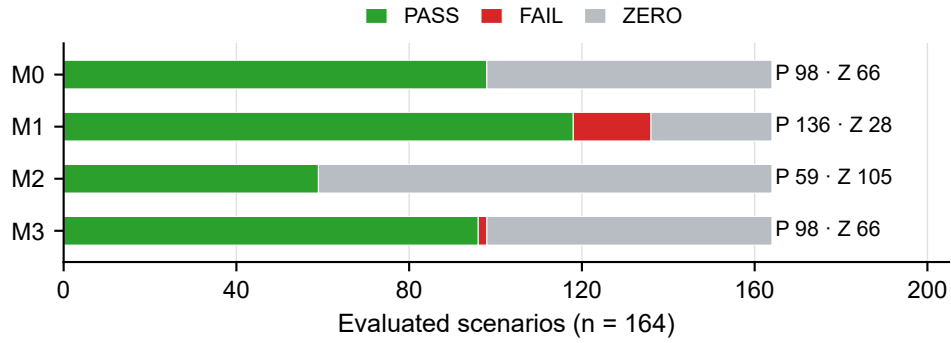
The access change had two parts. Both methods were positive in 98 scenarios. M0 passed 98 and failed 0; M1 passed 86 and failed 12. Their mean candidate flows on this common subset were 4.81 and 7.55 robots/min. M1 also admitted 38 scenarios where M0 returned zero. Of these additional opportunities, 32 passed and 6 failed. The margin withheld both usable access and access that failed the service criterion.

M0's zeros comprised 28 service-gate decisions and 38 decisions from margin subtraction and rounding. Both methods returned zero in 28 scenarios. Across their

(a) Reference construction from configuration-matched service tests



(b) Robot access and directly tested service



All positive recommendations directly tested; zero admission is not a service PASS.

Fig. 2. Pedestrian service and robot access. (a) Baseline and mixed-flow tests define an illustrative reference flow. (b) Outcomes for the four methods on 164 held-out scenarios: positive recommendations that pass or fail the service test, and zero admission.

respective positive subsets, mean candidate flow was 4.81 robots/min for M0 and 6.32 robots/min for M1. The shared-subset comparison separates lower offered rates from the change in which scenarios received access.

Median reference utilization increased from 16.7% under M0 to 75.0% under M1. Their intervals were [14.3, 22.5]% and [66.7, 80.0]%. The utilization difference was 58.3 percentage points, with interval [47.7, 63.3]. M0 exceeded 0 references and M1 exceeded 16. MAE decreased from 4.54 to 2.43 robots/min. Closer agreement with reference flows accompanied greater access, but also more failed direct tests.

Continuous speed measurements add context to the binary outcomes. The median of tested groups' median speed retention was 0.982 for M0 and 0.963 for M1. The lowest observed seed-level values were 0.883 and 0.071, respectively. The extreme zero-margin value came from a technically completed but severely degraded run. These summaries describe the positive subsets; passing the service test also depends on density and the completed-trip flow ratio.

Estimator Choice Changes the Allocation

The unconstrained width–demand estimator M3 produced aggregate outcomes close to M0. Both issued 98 positives and 66 zeros. M3 had 96 passes and 2 failures. Its utilization was 18.2%, compared with 16.7% for M0, and MAE was 4.55 versus 4.54 robots/min. The paired MAE difference was -0.012 robots/min, with interval [-0.043, 0.019]. Individual recommendations differed despite similar aggregate access.

The width-only method M2 offered substantially less access: 59 positives and 105 zeros. All positive recommendations passed. Median reference utilization was 0.0%, and MAE was 6.98 robots/min. M2 and M3 use their own margins, so these results compare complete allocation rules. Adding a free width exponent to M0 changed aggregate outcomes less than removing its margin; omitting pedestrian demand produced a more restrictive rule.

Transfer to Irregular Sidewalk Geometry

The Hong Kong experiments extended the analysis from idealized corridors to the polygon shapes in Fig. 3. Original configurations supported evaluation at 13 of 32 candidate sites (40.6%). Including the 5 entrance-repaired configurations increased this to 18 sites (56.2%). The increase depended on entrance placement within retained polygon shapes.

Table 3. Hong Kong admission outcomes, separated by entrance configuration.

Configuration	Method	Positive	Zero	PASS	FAIL	Exceed	U (%)
Original	M0	7	6	7	0	0	16.7
Original	M1	11	2	10	1	1	64.6
Original	M2	7	6	7	0	0	8.3
Original	M3	7	6	7	0	0	16.7
Repaired	M0	4	1	4	0	0	20.0
Repaired	M1	5	0	5	0	0	50.0
Repaired	M2	2	3	2	0	0	0.0
Repaired	M3	4	1	4	0	0	20.0

Original: 13 configurations, 10 positive references for U . Repaired: 5 configurations, 5 positive references for U . Exceedance uses every configuration in its stratum.

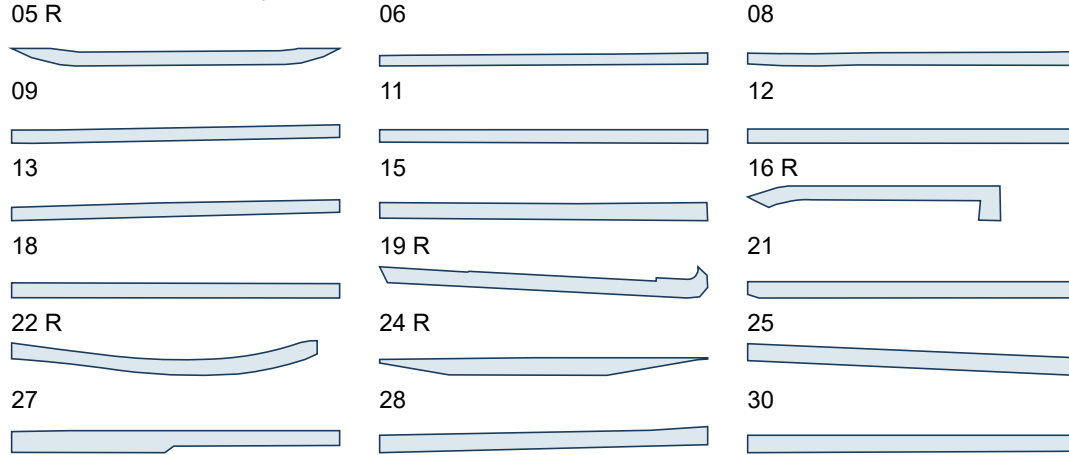
Among original configurations, M0 gave 7 positives: 7 passed and 0 failed. M1 gave 11 positives, including 10 passes and 1 failure. Their utilization was 16.7% and 64.6%. Among repaired configurations, all 4 M0 positives and all 5 M1 positives passed. M3 matched M0’s outcome counts in each stratum. M2 offered fewer positives in the repaired stratum. The pattern again combined the amount of access offered with configuration-specific service outcomes.

Sensitivity to Robot and Flow Assumptions

Robot dimensions and directional composition changed service outcomes at a fixed entry rate (Table 4). Two scenarios were selected by width-order quartiles before the sensitivity runs. At 4 robots/min, the 1.75 m scenario passed 30/30 seeds with the default

(a) Original polygon shapes

All final sites; individually scaled; R = repaired entrance



(b) Reference and recommended entry rates

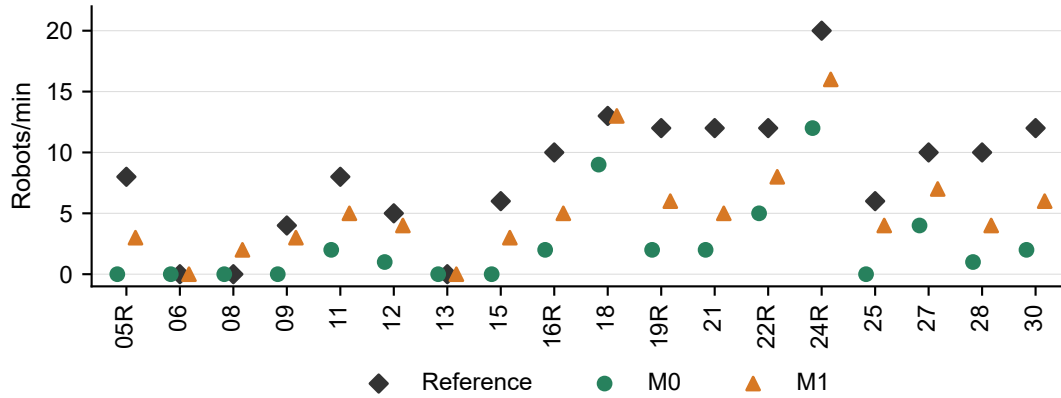


Fig. 3. Hong Kong geometry transfer. (a) Polygon shapes with independent display scaling and rotation; R marks an entrance-repaired configuration. (b) Reference flows and candidate entry rates using frozen development parameters. Original and repaired configurations are summarized separately in Table 3.

robot body and 0/30 with the larger body. The 2.4 m scenario retained 30/30 passes with the larger body. Changing the pedestrian directional split reduced passing seeds to 28/30 in the narrow scenario and 29/30 in the wider scenario.

Table 4. Sensitivity at 4 robots/min for scenarios of width 1.75 and 2.4 m. Each setting has a matching baseline.

Setting	Radius (m)	Narrow: pass/seeds	Wide: pass/seeds
Default, 50:50	0.480	30/30	30/30
Smaller body, 50:50	0.385	30/30	30/30
Larger body, 50:50	0.575	0/30	30/30
Default body, 75:25	0.480	28/30	29/30

The smaller and larger bodies use interaction radii 0.385 and 0.575 m. Estimators, margins, and service thresholds remain fixed in these checks. Their differing outcomes indicate that the recommendation is tied to robot geometry and pedestrian-flow composition.

Repeated seed blocks also changed service-test results near a service boundary (Table 5). In the preselected DC-063 case, a lower flow could fail when a higher flow passed. The pattern differed between original and independent blocks. This observed acceptance pattern does not establish a genuinely nonmonotonic population pass probability. It supports testing each candidate flow rather than inferring that every lower rate meets the criterion.

Table 5. Boundary case DC-063: passing seeds in original and independent blocks, each with its own baseline mean. Qualification requires 29/30.

Flow (robots/min)	Original: pass/seeds	Independent: pass/seeds
1	30/30	28/30
2	28/30	29/30
3	29/30	27/30
4	26/30	25/30

DISCUSSION

A margin changes the access that a sidewalk offers, not just the error of a fitted curve. It lowers candidate rates within shared admission scenarios and can remove access entirely in others. Some withheld opportunities would pass the service test. Choosing conservativeness involves weighing those lost opportunities against failed pedestrian-service recommendations. Reference error alone does not express that choice. The comparison with M3 suggests that an extra fitted coefficient offers less leverage over aggregate access than the margin in this experiment.

The framework makes these decisions operationally distinct. An out-of-domain case calls for further assessment or a different model. A service-gate zero points to pedestrian conditions or demand that are unsuitable for robot admission. A zero after margin subtraction reflects the allocation rule. A positive candidate specifies a flow for testing under the stated conditions. This structure complements operating permits and fleet limits by making the local service objective part of the entry-rate decision.

Geometry transfer adds an access-design requirement. A polygon can retain its original shape yet require a different entrance anchor for the simulated demand to enter appropriately. Managers applying a segment-level rule would need to specify where robots and pedestrians enter, as well as the width and demand used in the estimate. The Hong Kong results support this configuration-specific approach; they do not measure the coverage of all sidewalks in the city.

Several limits constrain transfer. Robots are represented as circular walking agents, without an independently calibrated yielding or obstacle-avoidance controller. The service criterion measures average walking conditions rather than collision risk or legal right-of-way compliance. Specified regular demand also limits interpretation under arrival

bursts or changing pedestrian activity. Completed-trip counts and finite observation windows differ from actual entry measurements, so nominal recommendations must remain distinct from realized throughput.

The finite flow scan and 30-seed groups provide evidence at tested conditions. They do not establish a continuous capacity interval or population pass probability. The bootstrap intervals describe site composition rather than all sources of uncertainty. Fixed gates were selected outside the city folds. Hong Kong is a simulation geometry transfer with prior site inspection, and field validation remains necessary. Robot-size and direction-split results identify two conditions that would require reassessment before applying a recommendation to a different operating setting.

CONCLUSION

Sidewalk robot admission can be framed as a local service–access decision by linking applicability, pedestrian baseline conditions, and candidate entry rates. On 164 held-out scenarios, removing the margin increased positive recommendations from 98 to 136 and observed failed tests from 0 to 18. The additional 38 opportunities included 32 passes and 6 failures. An unconstrained width–demand estimator changed aggregate outcomes less than removing the margin. Hong Kong transfer and sensitivity tests linked service outcomes to entrance configuration, robot size, and flow composition. The framework supports service-aware evaluation of candidate entry rates under stated simulation conditions, rather than defining a universal sidewalk capacity.

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